

- 3 (a) One way to measure global biodiversity is to count the number of species of organisms.

Table 3.1 shows estimates for 2009 of the number of species in some taxa of animals.

The numbers in brackets are the numbers that were updated in 2019 from the International Union for Conservation of Nature and Natural Resources (IUCN) for three intensively studied taxa.

Table 3.1

number of species in each taxon		
kingdom	phylum	class
animals 1 438 805	arthropods 1 191 770	arachnids 102 248
		crustaceans 47 000
		myriapods 16 072
		insects 1 024 945
		other arthropod classes 1505
	chordates 63 543	fish 31 269
		amphibians 6515 (6722)
		birds 9990 (11 126)
		reptiles 8734
		mammals 5487 (5692)
	molluscs 85 000	
	other animal phyla 98 492	

- (i) List **three** features shared by **animal** species.

- 1
- 2
- 3

- (ii) Explain how the classification of species into a taxonomic hierarchy assists the work of conservation bodies such as the IUCN.

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- (iii) Table 3.1 shows that the number of species of amphibians, birds and mammals has increased between 2009 and 2019.

Discuss whether the increase in numbers means that these classes of chordate are being successfully conserved.

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- (iv) With reference to Table 3.1, identify the class of animals that is most diverse then calculate the percentage of animal species that belongs to this class.

Show your working and give your answer to two significant figures.

answer = % [2]

(b) An analysis in 2019 concluded that twice as many insect species have populations that are decreasing in size compared with chordate species. This analysis focused on developed countries that have large human populations.

(i) Outline **two** factors that may cause populations of insects to decrease in size in developed countries that have large human populations.

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(ii) Explain how the introduction of crops that are genetically modified to express the Bt toxin can benefit biodiversity.

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[Total: 14]